The Association Between Upper Trapezius Activity and Thorax Movement in Classical Singing

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Summary: This study aimed to examine in classical singing the phasing of the activity in upper trapezius (TR) to upper and lower thorax movement and to the phasing of activity in the intercostals (INT) and in the lateral abdominal (OBL) muscles. Electromyographic (EMG) activity was recorded from the TR, INT, and OBL muscles on the right side. Thorax movement (TX) was traced with two strain gauge sensors placed around the upper and lower thorax. Four professional opera singers (soprano, mezzo, tenor, and baritone) and four advanced student classical singers (three sopranos and one mezzo) participated. Three of the professional singers were 33 years, and one was 40 years. The students were between 23 and 30 years. Different arias, freely chosen by the singers from their professional repertoire, served as the singing task for the opera singers. All students sang "Summertime" from *Porgy and Bess*. All subjects performed their task three times with variation in vocal loudness (normal, forte, piano). Thereafter, for all subjects, a biofeedback (BF) procedure was performed on TR to lower TR activity and a repeat performance of the singing tasks was carried out. EMG activity from the three recording sites and upper and lower TX circumference were compared before and after BF. A phasing of upper TR activity to INT and OBL activity was discovered, all muscles supporting the expiration phase. During phonation, the upper TR contributes in the compression of upper TX, thus serving as an accessory muscle of expiration. Group results from both opera singers and student singers showed that EMG activity was significantly lowered after BF. The lowered TR activity resulted in an expanded upper TX circumference and less TX respiratory movement after BF.

Key Words: Classical singing—Electromyography—Trapezius—Thorax—Biofeedback.

INTRODUCTION

Voice experts recognize the importance of breathing and its role as a pressure generator for voice production. Modifying the habits of respiration often solves a phonatory problem. However, a consensus understanding of the term "breathing support" in terms of muscle activation patterns is not properly established.[1] Furthermore, singers vary in their anthropometrics and posture and may, therefore, need individual adjustments in the use of the respiratory

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and postural muscles for the best singing result. Thomasson and Sundberg[2] found that the contribution to lung volume changes from thorax and abdominal wall movements varied across professional singers, thus suggesting that professional singing does not require uniform breathing strategy. Therefore, a likely reason why singing teachers teach breathing so differently is individual experiences from their own career.[3–5]

The detailed muscular characteristics of the singer's breathing pattern have not yet been fully explored,[6] even though the relevance of the respiratory apparatus to voice production has been intensively studied.[6–17] The breathing apparatus offers different strategies for inhalation and exhalation. When the rib cage has been expanded or compressed by muscular forces, it strives to return to its neutral volume, thus generating a nonmuscular passive expiratory or inspiratory recoil effect. With the body in an upright position, the abdominal content exerts a downward pulling force on the diaphragm, so that gravity has an inhalatory effect, counteracted by the abdominal wall. As the recoil forces change with the air contained in the lungs, the need for muscular forces to maintain the air pressure needed by classical singers during phonation, continuously changes with lung volumes. Thus, a complex system of passive recoil forces supplemented by active muscular forces would control subglottal pressure when classical singers produce the required pressure needed for the chosen vocal loudness and pitch.[1] The relevance of abdominal muscle activity to breathing support is described by Hixon et al[5]: "The abdominal wall is probably setting a platform off which the rib cage functions. The rib cage is probably the thing that is delivering the driving pressure."

The role of the abdominal muscles in the production of voice by classical singers has previously been investigated by electromyography (EMG) by Watson et al[17] and Sundberg et al,[6] who also included both the external and internal intercostals. Sundberg et al emphasized that their investigation analyzed only some of the breathing muscles and that it seemed to be a general opinion among singing teachers that the breathing muscles can be used in a number of ways, resulting in different phonatory characteristics. Therefore, in future research, the detailed muscular characteristics of these breathing patterns should be investigated.

In voice pedagogy, relaxed shoulders are frequently emphasized as desirable and good for breathing.[18,19] Nevertheless, a previous study with electromyographic (EMG) recordings showed that the upper trapezius (TR) can be activated to a relatively high level in classical singing by relatively untrained singing students.[20] The postural demands in the experimental situation seemed low, and we hypothesized that the activation of upper trapezius (TR) was influenced by other factors. Notably, the activity level tended to increase toward the end of strong exhalations, singing sustained tones to exhaustion. In an unpublished study, we also found this to happen by professional opera singers. A reduction in TR activity level caused little, if any, compensatory adjustment in the activation of the intercostals (INT) and the abdominal muscles (OBL) during singing. Therefore, to further search for a link between shoulder muscle activation and breathing, the present study aimed to investigate a possible TR influence on thorax (TX) movement in classical singing. Furthermore, we recorded EMG activity from the INT and the abdominal muscles (OBL) to determine any phasing of TR activity to INT and OBL activity. In this, EMG activity from the TR, INT, and OBL sites and upper and lower TX movement were recorded. Finally, we looked for possible effects of different TR activation levels on the classical singer's use of TX. In this, TR activity was lowered by use of EMG biofeedback (BF) and TX movement before and after BF was compared.

MATERIAL AND METHODS

Subjects

Eight singers (six females and two males) participated in the study. Four professional opera singers constituted the opera group (one soprano, one mezzo-soprano, one tenor, and one baritone). Four advanced students in the third and fourth year of their Conservatory study constituted the student group (three sopranos and one mezzo) (Table 1).

Recordings

Muscle activity was recorded from the TR, INT, and OBL by use of an ambulatory four-channel monitoring system (Physiometer PHY 400, Premed,

TABLE 1. *Voice Type and Age on All Singers Participating in the Study*

Subjects	Voice Type	Age
Opera singers		
	Soprano	33
	Mezzo soprano	33
	Tenor	33
	Baritone	40
Student singers		
	Soprano	23
	Soprano	24
	Soprano	30
	Mezzo soprano	23

Norway). Silver/silver chloride bipolar electrodes with an active diameter of 6 mm and a center-to-center distance of 20 mm were used to record the surface EMG signal. The EMG signals were band-pass filtered at 20–800 Hz and sampled at 1600 Hz. They were, thereafter, A/D converted, and the root-mean-square (RMS) value was calculated and transmitted at 10 Hz on a serial interface to a PC laptop. The processed signals were further analyzed by use of the Physiometer software, using a time resolution of 0.2 s. The electrode positions are indicated in Figure 1. For the TR, the medial electrode was placed 2.5 cm lateral to the midpoint of a line from the spine of the C7 vertebra to acromion. For OBL, the rostral electrode was placed 2 cm below the lowest part of the rib and the lower electrode 2 cm further down toward the os pubis. For the INT, the caudal electrode was placed between costae 9 and 10, perpendicular down from papilla mammaria, and the rostral electrode was placed between costae 8 and 9 on the same line. The ground electrode was placed on the spine of the C7 vertebra.

The EMG responses were calibrated in percent of the EMG response in maximal voluntary contraction (%EMG_{max}), using static contractions with resisted movement in the muscle force direction. TR calibration was performed by arm elevation with the arms abducted 90° in the scapular plane. A maximal compressed and elevated rib cage served as movement pattern for the INT, and for the OBL, the attempt to maximally compress the abdomen gave the best response. Three attempts of a maximal contraction

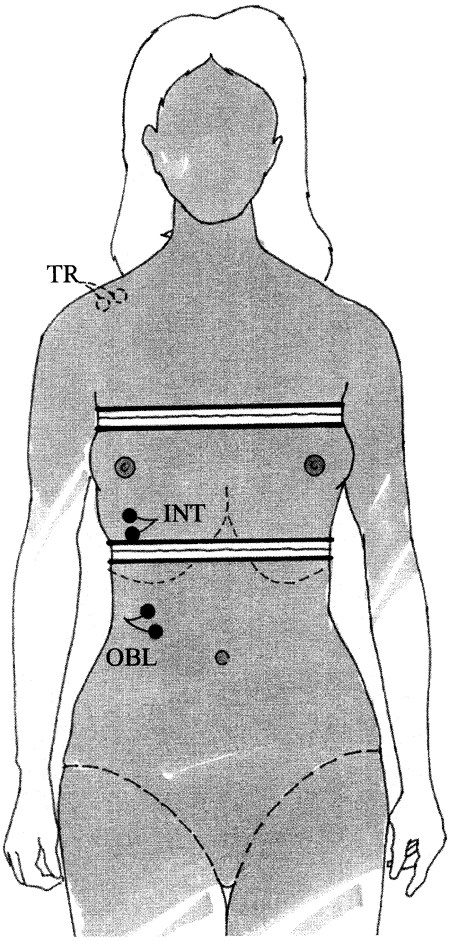

FIGURE 1. The anatomical position of the electrode placement on upper trapezius (TR), intercostals (INT), and oblique (OBL) muscles and the position of respiration sensors on thorax (TX) are shown.

were made for each muscle, and highest voltage recorded was used.

The mean and range values of the RMS-detected calibration responses were for TR 730 μV (350–1573 μV), INT 128 μV (88–225 μV), and OBL 110 μV (55–190 μV). Thus, the EMG response was six times higher for TR than for the other muscles with the maximal contractions calibrated in technical units (μV). The large range of maximal EMG responses for each of the muscles is not unreasonable in view of very different body mass indexes (ie, weight divided by height squared) of the singers, which significantly influence the EMG signal amplitude. A source of error is that the singers may not have

performed their maximal exertion with the highest effort. Inspection of the μV values and our previous experience with the calibration procedure makes us believe the responses for the TR are fair representations of maximal force exertions. The most problematic muscles in this respect were the OBL and INT, where no external lever is available. Some singers produced higher response during singing than in the calibration exertions, and the reference values were adjusted to accept these new values as maximal responses.

TX movement pattern was traced with two strain gauge sensors (RES-117) placed around the TX and sensing thorax (TX) circumference. One sensor was located with the upper edge at the axilla level, and the other was located around the lowest part of the TX, as illustrated in Figure 1. The sensors were tested and found to show a linear relationship between transducer output and chest circumference within the movement range defined by maximum expansion and maximum contraction of TX in regular singing. Transducer output is dependent on the relative movement range (ie, the movement is registered larger for the same expansion of the chest wall, measured in centimeters, if the circumference is small). TX movement is, therefore, not easily compared between subjects, but within-subject changes in chest movement with the respiratory cycle and the effect of altered TR activity on chest movement can be assessed.

Procedure

Different arias, freely chosen by the singers from their professional repertoire, served as the singing task for the opera singers. All student singers sang "Summertime" from *Porgy and Bess*, rehearsed the day before the recording. Both opera and student singers performed the singing task three times with variation in vocal loudness (normal, forte, piano). The singers were asked to envision the normal version as singing in a "normal" sized concert hall. The forte version was to be envisioned sung as if performed in the largest concert hall they had performed in, and the piano version in the smallest concert hall they had performed in. This instruction was given to create as much performance environment as possible in the singing procedure. Thus, it was the choice of the individual singer to choose vocal loudness in accordance to what they were used to when performing. However, it is not possible to create an experimental procedure to be exactly like a performance. Therefore, the subjectivity in the judgment of what was required from each singer has to be acknowledged.

The three versions of the aria were repeated after the BF session for a total of six separate recordings. For all subjects, the electrodes were attached during the whole recording session. The BF procedure was carried out by use of a PC laptop monitor, and it lasted approximately 10 minutes for each subject. The same PC laptop was used to monitor the EMG signals.[20] During the BF session, the singer looked at the monitor and observed the recorded muscle activity from the TR. By looking at the monitor while varying the activity level, the singer experienced that he/she could influence the level of activity by varying the muscular effort. The first author of this paper then instructed the singer how to minimize the TR activity level. The singer did not observe the monitor during the singing sessions before and after BF. Before the three recording sessions after BF, they were instructed to perform exactly as they did in the first three arias. In addition, singers were asked to focus on keeping the activity level of TR as low as possible by using their experiences from the BF session. The opera singers were recorded during an intensive period of rehearsals and performances. Each recording was, therefore, interrupted after 1–1.5 min to avoid six revisions of the entire aria. The same procedure was followed when recording from the student singers. The entire recording session lasted approximately 60 minutes for each subject, including the mounting of electrodes and respiration sensors, calibration maneuvers, 10 minutes of biofeedback, and recordings.

Analysis

The recordings were transferred as ASCII files to a PC for further analyses, and all data from all subjects were used in the analyzes. The data were statistically tested by use of SPSS 11.0 for Windows (SPSS, Chicago, IL). The RMS-detected EMG activity and chest transducer responses, both quantified at a time resolution of 0.1 s, were used to calculate the Spearman's rho (ρ) correlation coefficient when testing for associations between the EMG recordings

and with the TX movement pattern. The comparison of responses before and after BF was tested by Wilcoxon nonparametric two-related-samples test both for the TR and TX activity. The analysis of responses was carried out without blinding of the researchers; however, this should not affect the outcome because the recording and subsequent analysis procedures were not influenced by subjective judgment and no recording periods were eliminated.

Results

Time plots of EMG responses and TX movement of a female subject, representative for the majority of singers in this study, are shown in Figure 2. INT, OBL, and TR activity show clear expiratory phasing (end of expiration marked by arrows). INT and OBL activity is inversely correlated to lower TX expansion, and TR activity is inversely correlated to upper TX expansion. The activity patterns in Figure 2 were quantified by calculation of correlation coefficients between channels: TR versus INT has a correlation coefficient (ρ) of 0.28, INT versus OBL: $\rho = 0.71$, TR versus upper TX: $\rho = -0.47$, and OBL versus lower TX: $\rho = -0.31$. A recording with deviant activity pattern, presenting short inspiratory peaks in the OBL position, is shown in Figure 3. This gives a positive correlation for the OBL with lower TX expansion ($\rho = 0.31$); however, the correlation coefficients for TR versus INT ($\rho = 0.35$) and INT versus OBL ($\rho = 0.82$) were similar to those found for the recording in Figure 2. The correlation coefficient for TR versus upper TX ($\rho = -0.1$) is not significant. As OBL acts to compress the rib cage and has no inspiratory function, the OBL electrode for this person may have also recorded activity from the diaphragm. Weak inspiratory transients are seen in the INT recording, indicating either a low level of inspiratory activity by the external intercostals or cross talk from the diaphragm.

Figure 4 shows a summary of the correlation results. Each data point presents the mean correlation of singing the three versions of the aria, shown separately for recordings before and after BF. The correlations for the individual performances varied for each singer within the range defined by the corresponding group results in Figure 4; eg, the correlations of TR versus INT varied between -0.1 and 0.4 for all 48 individual performances. TR versus upper TX showed significant negative correlation for 36 (75%) and negative correlation for 42 (88%) of the 48 individual recordings. The OBL and INT showed the highest correlation. TR was for most subjects positively correlated with the INT and OBL. All three muscles were negatively correlated with chest circumference. These associations were unchanged after BF. Thus, TR, together with INT and OBL, supports exhalation in classical singing. TR activity decreased during the inhalation phase in most recordings, sometimes with complete inhibition of activity during the first seconds of a phrase.

All singers reduced their TR activity level on repeated singing after the BF session, as expected from a previous study.[20] Time plots of TR activity (Figure 5), upper TX (Figure 6), and lower TX (Figure 7) movement by the four opera singers illustrate a range of results regarding changes in TR activation and chest movement after BF. The singer in Figure 5A showed a marked reduction in TR activity after BF. The respiratory movement of the upper TX was mostly eliminated, except for brief inspiratory expansions (Figure 6A). Lower TX showed respiratory movement at a reduced level (Figure 7A). Contrasting this result, but consistent with a habitual respiratory role for TR in singing, only minimal change of TR activity pattern and respiratory TX movement after BF was observed for the singer in Figures 5D, 6D, and 7D. Intermediate to these two extremes in their response to BF, the singers in panels B and C of Figures 5–7 lowered their TR activity and reduced the respiratory chest movements, for the upper TX in particular. The subject in panels C (Figures 5–7) after BF started singing with very low TR activity that increased after 15 s. At the same time, upper TX movement pattern changed from being relatively still to show more respiratory movement. Table 2 presents the overall results for the effect of BF on the TR EMG activity and the chest movement in singing the aria. Subgroup analyzes results from the opera singers, and the student singers were also performed. Except for range upper TX for the students ($p = 0.07$), both groups showed significant effects of BF, corresponding to the results in Table 2.

After BF, the singers kept the upper TX in a more expanded position, with a reduction in the range of

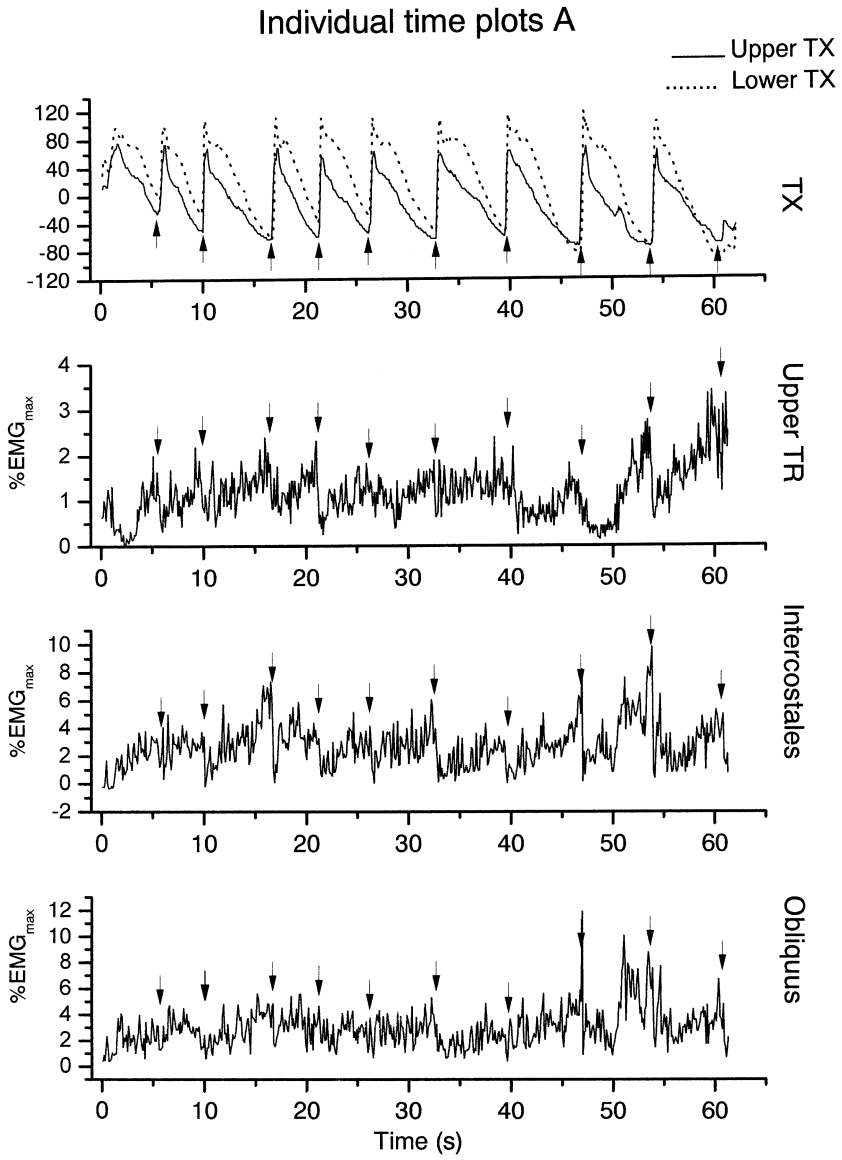

FIGURE 2. Individual plots of TR, INT, and OBL sites EMG activity, and upper and lower TX movement (female singer before BF, normal version of aria). Vertical arrows indicate end of expiration.

upper TX movement. The change in chest circumference represents a 10% expansion of upper TX and a 9% reduction of lower TX, on average. This pattern of change can be observed in the recording of subject C (Figures 5 and 6) after BF: With low TR activity (less than 0.5 % EMG_{max}), the first 14 s of the recording, an expanded upper TX position is found in the same part of the recording.

DISCUSSION

The main new finding in this study is that TR supports exhalation by the classical singers participating in this study. The duration of the singing sequence represented 10 or more respiratory cycles and revealed a habitual way of body use. The high resolution of the recordings makes the number of data points entered in the correlation analyses high

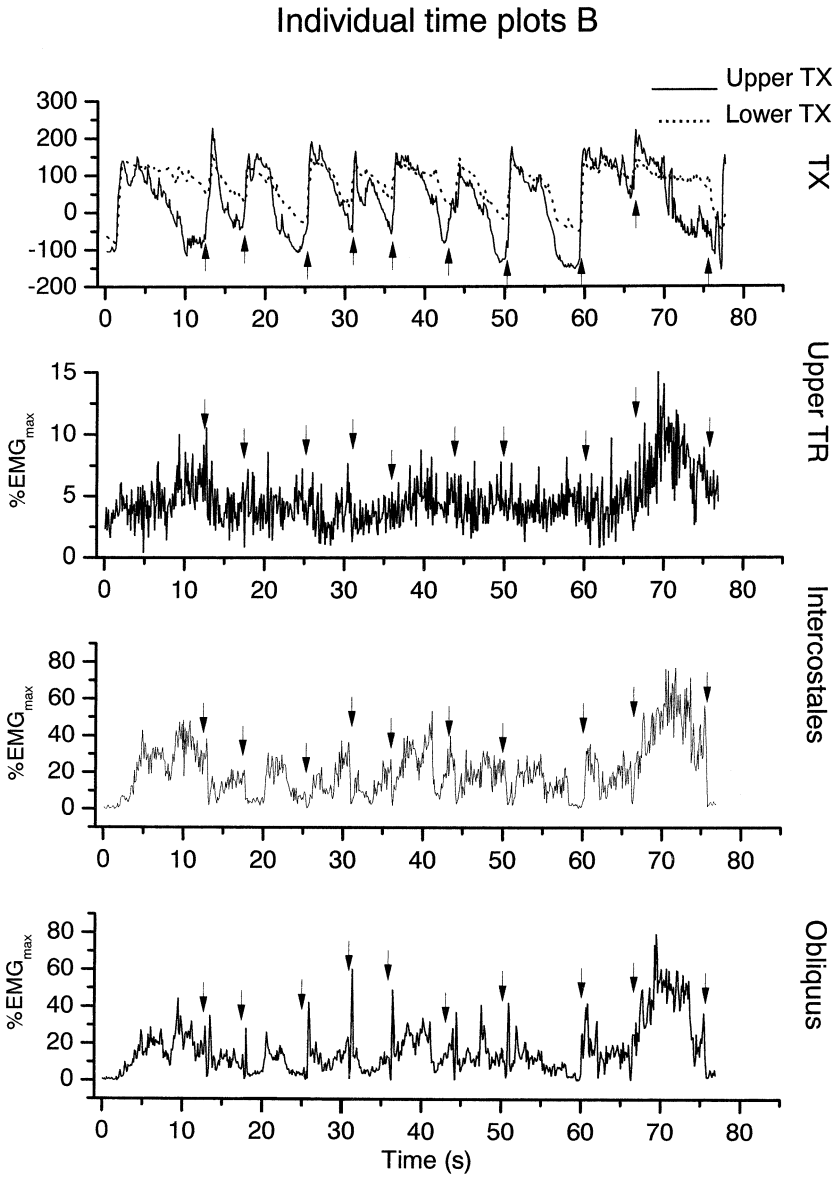

FIGURE 3. Individual plot of TR, INT, and OBL sites EMG activity, and upper and lower TX movement (male singer before BF, normal version of aria). Vertical arrows indicate end of expiration.

(600–1000), thereby reaching statistical significance for low correlations. However, visual inspection of the EMG recordings confirms that the TR, INT, and OBL support exhalation and have a consistent activity pattern for the individual breaths, ensuring that the results are trustworthy.

A cautionary remark is that the activity patterns of large muscles are monitored by one EMG electrode with limited pick-up area. We, therefore, do not know the extent the EMG recordings are representative for the activity pattern of the whole muscle. Cross talk of signals between muscles is another methodological concern. This error is difficult to estimate; however, the TR is an extensive and fairly thick muscle at the recording site with a well-defined fascia underneath. Previous studies have shown that the electrode placement of the TR records a representative activity pattern for upper

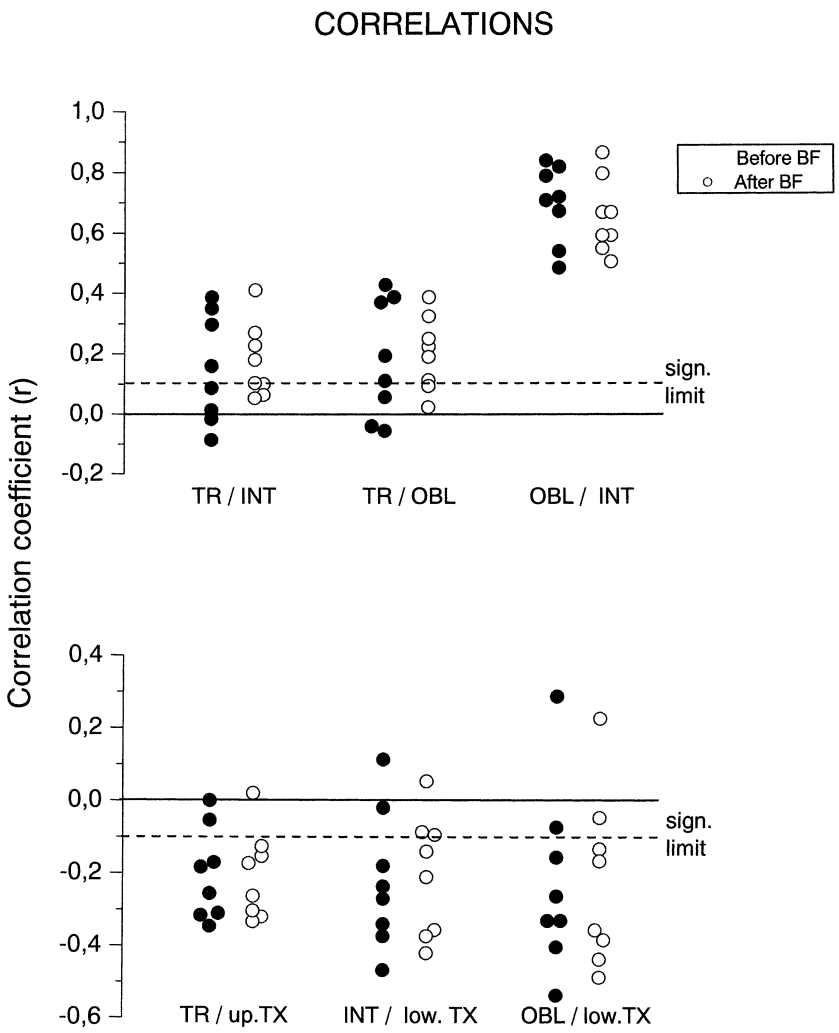

FIGURE 4. Results of correlation tests (Spearman ρ) between recordings. Each data point represents the mean correlation of the three versions of the aria by one singer. Results before (filled circles) and after (open circles) BF are shown.

TR.[21,22] The EMG signal from the TR is, therefore, unlikely to be significantly contaminated by the underlying supraspinatus muscle. Likewise, the INT electrodes may record contributions from both the external and the internal intercostals. The INT recording was dominated by expiratory phased activity, indicating that the internal intercostals were the most active muscle at the INT site. Sears[4] found that with increasing demand for sound intensity, the inspiratory muscles became inactive and the expiratory muscles became active at higher lung volume. It should also be kept in mind that the INT recording is not representative of activation of the full rib cage, which may show differential activity pattern, depending on lung volume.[9] The OBL recording likely has contributions from both the internal and external oblique muscles and the traverses abdominis. It is difficult to assess the proportion of activity from the different muscles, which may change depending on the need for control of the subglottal pressure.[23,24]

The contribution of the TR to exhalation is habitual and not necessary in a respiratory mechanics sense, as this and the previous study[20] demonstrate that the singing task is executed without problems with markedly reduced TR activity after BF. One

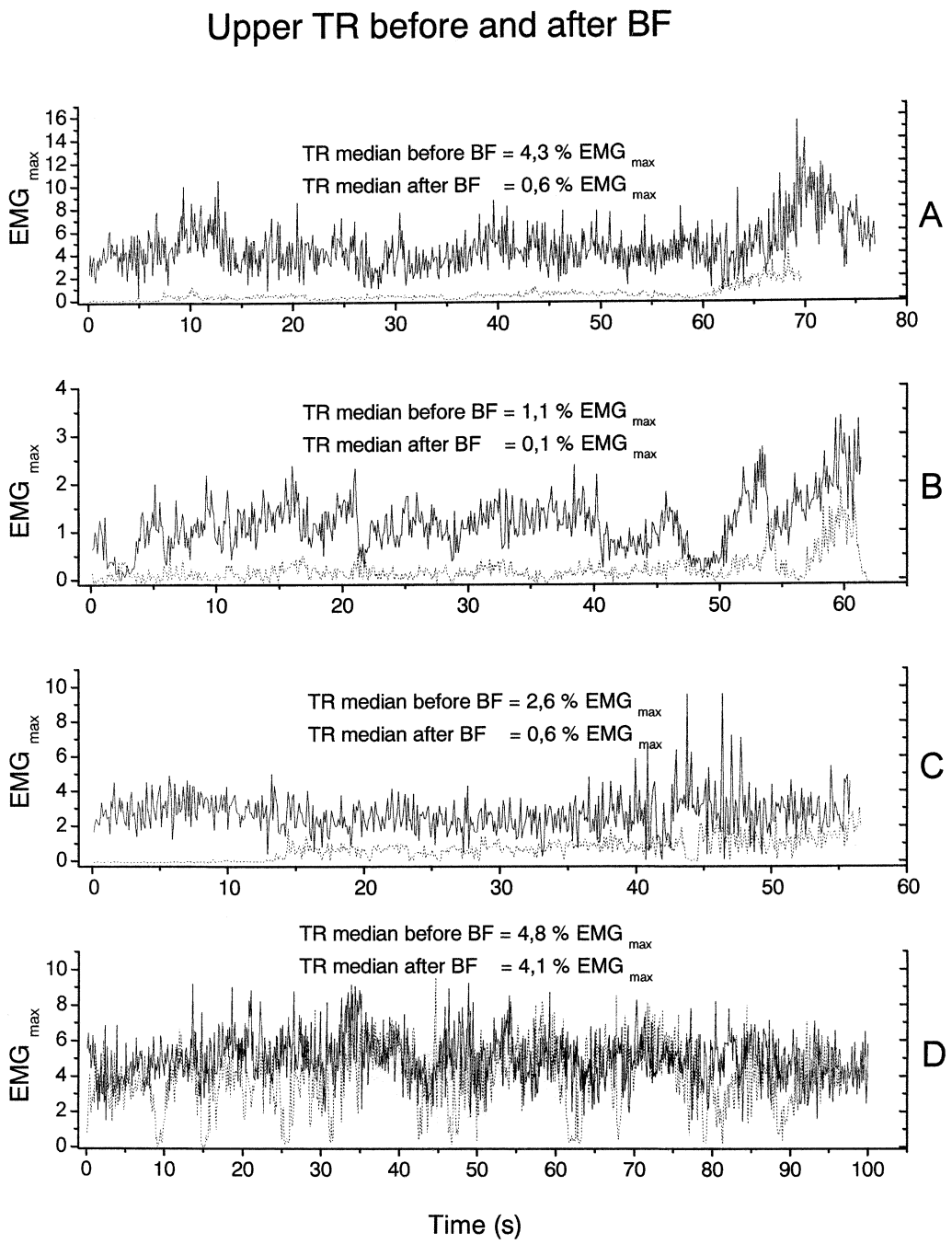

FIGURE 5. Time plots of TR site EMG activity by the four opera singers in the normal version of the aria, before and after BF. Dashed line is after BF.

of the professional singers presented a low activation level of TR before BF, despite singing in a very powerful vocal loudness. Upper TX is kept in a "higher" (ie, more expanded) position when the TR activity is reduced. With unchanged head posture, a reduction in the TR activity lowers the upward pull on the clavicle and reduces the medial rotation of scapula, presumably accounting for a more

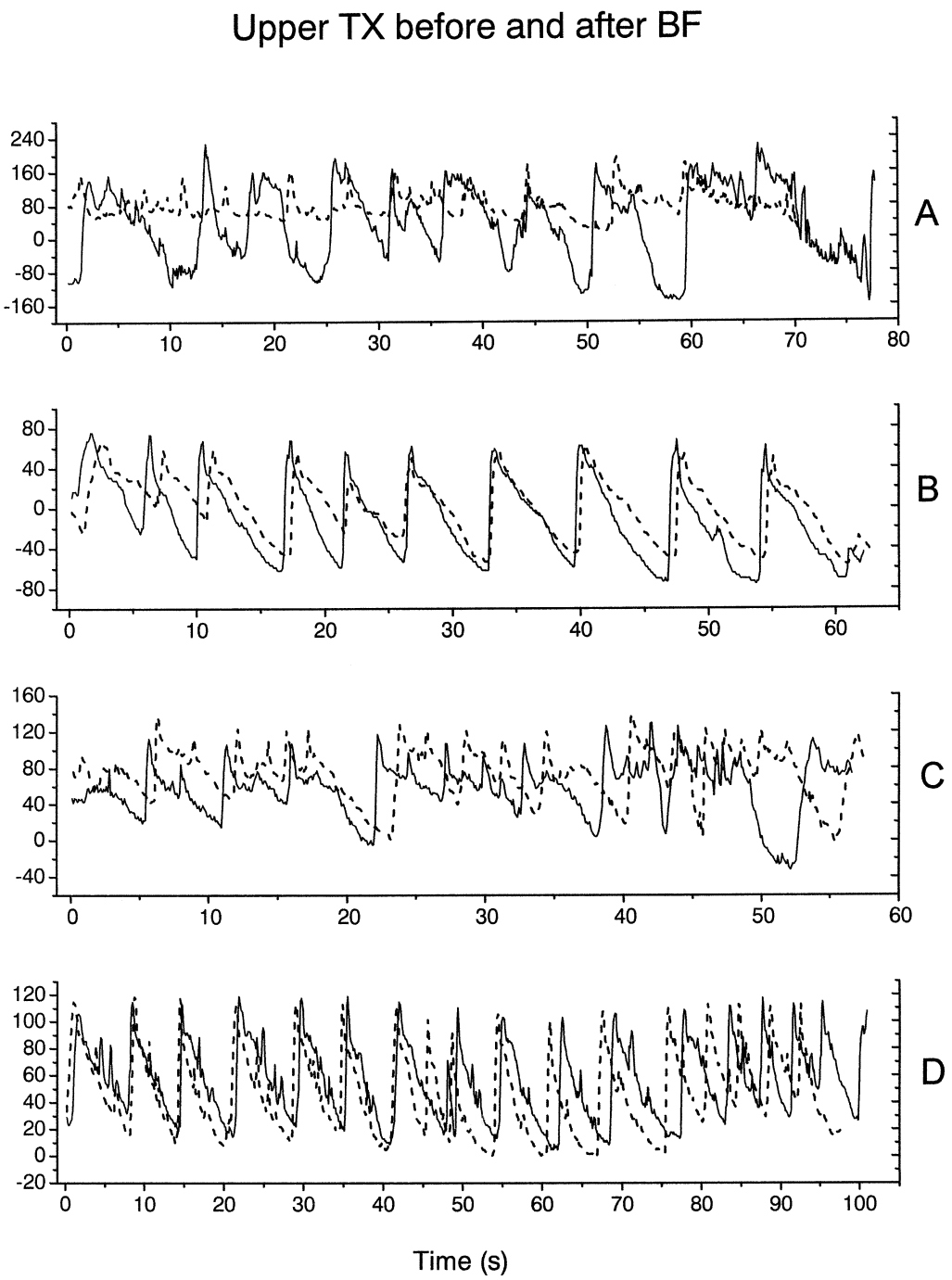

FIGURE 6. Time plots of upper TX moving pattern by the four opera singers in the normal version of the aria, before and after BF. Dashed line is after BF. Same recordings as in Figure 5.

expanded upper TX. The reduction in movement range of upper TX implies that the reduction in TR activity leads to less separation of maximum inhalation and exhalation positions. This change in breathing pattern should be favorable to singers, according to Hixon et al's statement[5]: "Trained singers would avoid the extremes on the relative-volume chart. They would stay more centralized on their chart. They

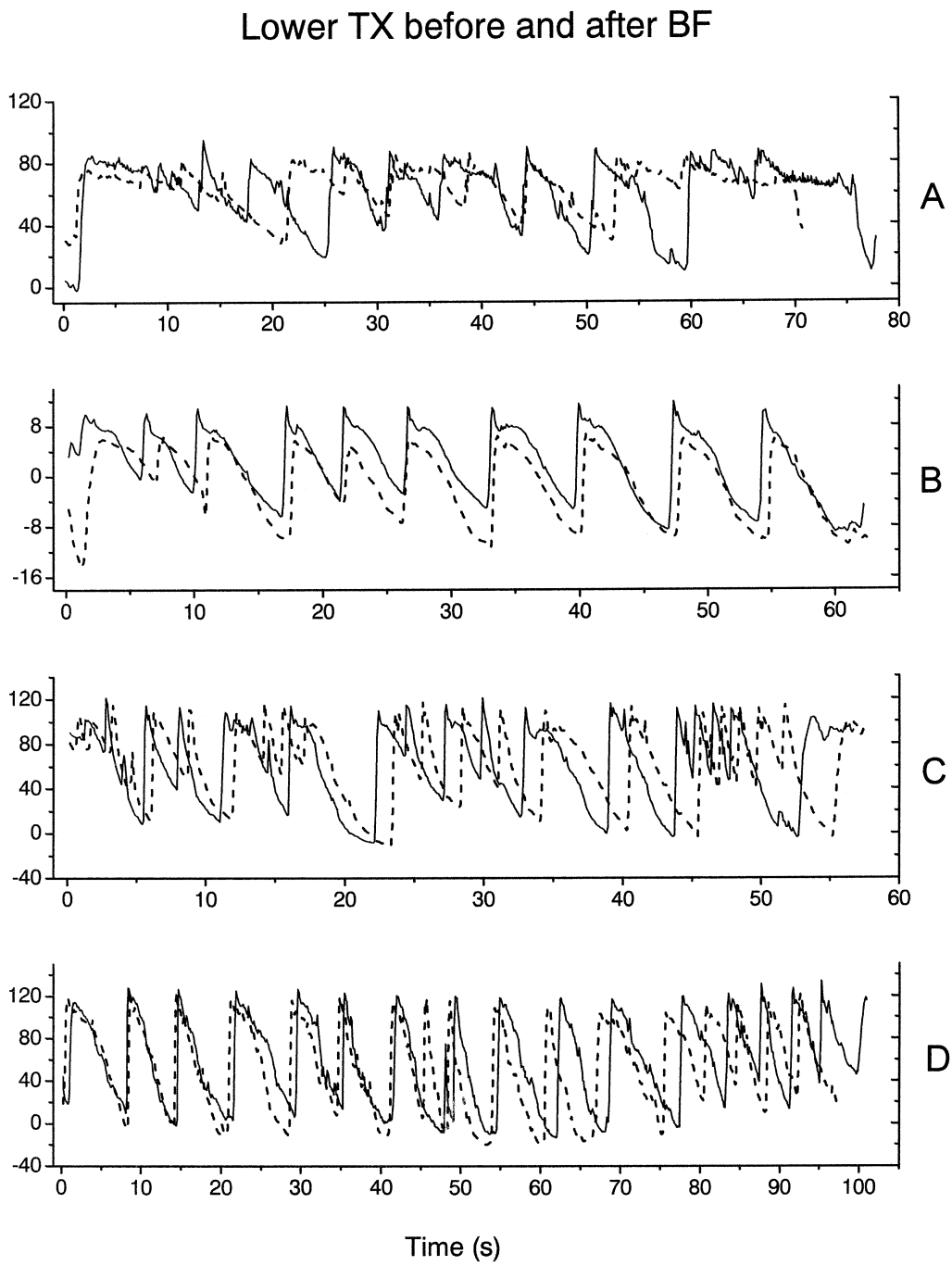

FIGURE 7. Time plots of lower TX moving pattern by the four opera singers in the normal version of the aria, before and after BF. Dashed line is after BF. Same recordings as in Figure 5.

would not go into extreme positions in controlling any of the chest wall parts. That would be the most economical biomechanical strategy."

All singing procedures were carried out in an upright posture, with arms hanging down. Postural demands would, therefore, be slight, and the TR, INT, and OBL should predominantly contribute to generate subglottal pressure. With the reduction in TR activity after BF, other exhalation mechanics must replace the TR contribution. The dynamic

TABLE 2. *Statistical Analyses of Paired Differences in TR and Chest Movements for Opera and Student Singers (before/after BF)*

	Median difference, before/after BF		Test statistic Z	Sig. (Two-tailed)
	EMGmax	Arbitrary units		
Upper TR				
Difference in means	1.2		−4.1	0.000
Upper TX				
Difference in means		20.8	3.4	0.001
Difference in range		−31.3	−3.1	0.002
Lower TX				
Difference in means		−12.8	−3.3	0.001
Difference in range		0.5	−0.9	0.345

The statistics are calculated on the basis of individual performances of the aria (n = 24/24). Mean differences are shown.

levels of singing were unchanged after BF, and a certain pitch and dynamic level would normally require the same subglottal pressure. A possible contributor in maintaining subglottal pressure is a more active lower chest wall; however, INT and OBL activity did not change significantly with the TR activated at two different levels. Conceivably, a higher and freer upper TX may allow more action in lower TX, without significantly changing the activity level of muscles in the lower chest and abdomen that support exhalation.

Alternatively, a more expanded upper chest may lead to a lowered larynx, thus changing the settings for the needed subglottal pressure. Iwarsson[25] found in a study of untrained singers that an elevated rib cage was associated with a lower laryngeal position than with a reduced rib cage volume. Hoit[8] stated that an expanded rib cage is advantageous for classical singing, in which the demands for rapid and precise pressure control are even greater than for speaking. In fact, classical singers have been shown to employ very large displacements of the abdomen to elevate the rib cage to a high position during performance.

Most singers presented muscle activity patterns that supported our main conclusions, illustrated by the time plot in Figure 2. Nevertheless, individual variations were seen, as in the time plot in Figure 3. In this recording the correlations of lower TX versus INT ($\rho = 0.08$) and lower TX versus OBL ($\rho = 0.35$) were positive. This is contrary to the results by all other singers, and it implies that lower TX circumference increases when the INT and OBL sites activity increases. Inspection of the activity patterns of the TR, INT, and OBL sites indicates several different aspects in this recording. A similar activity pattern among the three muscles is seen during the ending of the aria sequence (60–75 s). TX is kept well expanded during the final period, including a small, quick inhalation at 66 s. The EMG activity of all muscles is much higher than earlier in the sequence, and the expanded lower ribs indicate recruitment of inhalation forces. The balancing of the lower TX could be a muscular interplay between the diaphragm on the one hand and the three ventral muscles on the other. Cresswell et al[26] and Goldman et al[27] have both noted that the superficial and deep muscles of the ventrolateral abdominal wall tended to act in concert during voluntary breathing efforts. OBL site EMG activity, showing much higher values than earlier, may record both obliques and transvesus abdominis in this sequence.

The statement of Shirlee Emmons[18] and other voice experts[28] on keeping the shoulders relaxed and independent of the breathing mechanism to improve breath control in classical singing may well be referring to the TR role as a pressure generator, as found in this study. We assume that the TR contribution in creating a subglottal pressure is not a conscious act of the singer in composing the breathing support. It still is thought provoking that all singers in this investigation use the TR this way. Individual results before BF and most of the results after BF show that singing can be performed with TR activated to a low level. This fact, and the consequences of high TR activation on TX moving pattern, should be communicated to singers in teaching correct posture and breathing behavior.

CONCLUSION

A phasing of upper TR activity to INT and OBL activity was discovered, all muscles supporting the expiration phase. During phonation, the upper TR contributes in the compression of the upper TX, thus serving as an accessory muscle of expiration. EMG activity was significantly lowered after BF, and lowered TR activity resulted in an expanded upper TX circumference and less TX respiratory movement after BF.

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